

Foundations of Biomedical Informatics
Quiz - 1

Date - 23 January 2025

Time - 30 minutes

Name:

Multiple Choice Questions (MCQs) 4*10 = 40

1. You discover a previously unnoticed bias in your ICU outcome model favoring younger patients. Which ethical principle is most relevant to address this?
 - a)Autonomy
 - b)Non-maleficence
 - c) Beneficence
 - d) Justice**
2. If you notice that MIMIC lab data has missing values predominantly in older patients, what is the implication for predictive modeling?
 - a) The model will automatically adjust
 - b) Older patients can be removed without consequence
 - c) It may introduce systematic bias if missingness correlates with age**
 - d) It is safe to impute with the mean
3. A collaborator asks for access to your MIMIC-derived dataset for exploratory analysis. Which action aligns with ethical research practice?
 - a) Only provide summary statistics
 - b) Require completion of CITI training and a Data Use Agreement before sharing**
 - c) Share the data freely since it is de-identified
 - d) Share only after publishing the results
4. During secondary research, you identify patterns suggesting disparities in care among demographic groups. What is the most ethically responsible approach?
 - a) Aggregate data to hide demographic differences
 - b)Remove minority groups to improve model performance
 - c) Ignore disparities to avoid controversy
 - d) Report disparities transparently while protecting privacy**
5. Why is it necessary to document assumptions and limitations when publishing research based on MIMIC data?
 - a. It is only required for grant applications
 - b. It prevents others from using your dataset
 - c. It increases journal acceptance chances automatically
 - d. It ensures transparency, reproducibility, and ethical integrity**
6. Why must a Data Use Agreement (DUA) specify permitted analyses even for de-identified data?
 - a. To improve computational efficiency
 - b.Because de-identification removes all legal obligations
 - c. To prevent misuse or attempts at re-identification**
 - d.To comply with journal formatting rules
7. A researcher is analyzing ICU data to study a new drug. The researcher owns stock in the pharmaceutical company producing the drug. What is the primary ethical concern? a. IRB

approval

b. Data privacy

c. Conflict of interest

d. Statistical power

8. A junior researcher wants to publish ICU data analysis without proper CITI training. Which principle is at risk?

a. Respect for persons

b. Beneficence

c. Justice

d. Non-maleficence

9. Which practice demonstrates integrity in reporting research findings?

a. Omitting subgroup analyses that show poor model performance

b. Modifying data to improve outcomes

c. Documenting limitations, bias, and assumptions transparently

d. Selectively reporting only statistically significant results

10. You are conducting a multi-center ICU study. One site reports data inconsistently. Which ethical principle should guide your action?

a. Autonomy

b. Non-maleficence

c. Beneficence

d. Justice

Short-Answer Questions

1. You are conducting a study on ICU readmission outcomes, funded by a pharmaceutical company. During model development, you discover that your predictive model systematically underestimates risk for patients from a minority ethnic group.

a) Identify the conflict of interest (COI) present in this scenario. **(1)**

ans. Financial conflict of interest due to pharmaceutical company funding influencing research outcomes.

b) Explain how you would manage or disclose this COI to maintain research integrity and comply with CITI training principles. **(1)**

Ans Disclose funding sources and potential COI transparently, follow institutional policies and ensure independent validation as per CITI guidelines.

c) Which ethical principles are relevant to addressing the model's bias? **(1)**

Ans. Justice (fairness across groups),

Non-maleficence (avoiding harm due to biased predictions)

Beneficence (improving care outcomes).

d) Explain HIPAA. **(2)**

Ans. HIPAA (Health Insurance Portability and Accountability Act) is a U.S. law that protects patient health information by regulating its use, disclosure, and safeguarding, including rules for de-identification and permitted research use.

2. You want to share a de-identified ICU dataset with a collaborator at another institution. **(1*5)**

a) Should you share the dataset? Explain your answer with reference to ethical and legal

considerations.

Ans. Yes, conditionally. Data can be shared only if it is truly de-identified and the use aligns with HIPAA, institutional policies, and ethical principles of respect for persons and non-maleficence.

b) What ethical steps must you take before sharing the data?

Ans Confirm de-identification, define a legitimate research purpose, ensure collaborator accountability, and minimize risk of patient harm or misuse.

c) What legal steps must you follow (eg. Data Use Agreement, HIPAA compliance)?

Ans. Sign a Data Use Agreement (DUA) defining scope of use, prohibit re-identification, and ensure HIPAA-compliant de-identification standards are met..

d) How would you ensure that the collaborator complies with HIPAA and CITI training requirements?

Ans Verify mandatory CITI certification, restrict access via institutional credentials, and enforce compliance through the DUA.

e) Suggest two strategies to minimize the risk of re-identification of patients in the dataset.

Ans **Remove or deidentify identifying fields**, such as exact dates, locations, so individual patients cannot be uniquely identified.

Limit data access and prevent dataset linking, ensuring the data cannot be combined with external sources to trace back to individuals.